Quantification: 877_wt_z073c1-2
==============================
PRE-VALIDATION: no number here has been checked against manual (ImageJ) counts.
PIXEL UNITS: lengths px, areas px^2, densities px^-1. The images carry no micron calibration.
Statistics are over interior lacunae only (those not touching the frame edge).

Per image
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measure	value
lacuna count	11
interior lacuna count	10
frame-edge lacuna count	1

Interior lacunae only: mean, median and sample SD (n = 10)
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measure	kind	mean	median	sd	units
area_px2	lacuna shape	3052.5	2716	1789.94	px^2
major_axis_length_px	lacuna shape	98.8449	103.026	32.5447	px
minor_axis_length_px	lacuna shape	40.3072	40.5486	15.7541	px
aspect_ratio	lacuna shape	2.628	2.2321	1.0659	unitless
eccentricity	lacuna shape	0.8894	0.8935	0.0708	unitless
solidity	lacuna shape	0.8013	0.8097	0.0523	unitless
roots_count	headline	12.3	11	7.6746	count
ring_length_r30_px	headline	453.9	426.5	213.64	px
ring_length_r60_px	reported	1032.3	1007	422.499	px
owned_length_px	ownership-dependent	2142.82	1744.59	1614.1	px
edge_count	ownership-dependent	107.6	83	84.4738	count
mean_edge_length_px	ownership-dependent	20.5446	19.6969	2.1018	px
perimeter_px	normalised	310.044	299.676	105.33	px
ring_area_r30_px2	normalised	10426.7	10232.5	2662.66	px^2
ring_area_r60_px2	normalised	25344.7	25657	5710.76	px^2
in_frame_fraction_r30	normalised	0.9859	1	0.0445	unitless
in_frame_fraction_r60	normalised	0.9634	1	0.0826	unitless
ring_density_r30	normalised	0.0419035	0.0450415	0.0132976	px^-1
ring_density_r60	normalised	0.0394457	0.0411739	0.0100114	px^-1
roots_per_100px_perimeter	normalised	3.6197	4.1631	1.7533	per 100 px
ring_attached_length_r30_px	network v2	402	404	219.901	px
ring_attached_length_r60_px	network v2	836.6	801.5	487.523	px
ring_length_w_r30_px	network v2	519.39	498.538	243.65	px
ring_length_w_r60_px	network v2	1186.83	1162.93	488.329	px
sholl_crossings_r10	network v2	10.8	10	3.8528	count
sholl_crossings_r20	network v2	12.4	11.5	6.8993	count
sholl_crossings_r30	network v2	16	16.5	6.1464	count
loop_count	loops, report only	10.6	7.5	11.296	count
loop_edge_share	loops, report only	0.2847	0.259	0.1648	share
ring_width_mean_r30_px	width	6.6029	6.8491	0.7424	px

Per field
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measure	value	units
analysed_area_px2	1.01731e+06	px^2
total_skeleton_length_px	31982	px
canalicular_length_density_per_px	0.0314378	px^-1
junction_count	696	count
junction_density_per_px2	0.00068416	px^-2
lacuna_count	11	count

lacunae_per_px2	1.081e-05	px^-2
skeleton_component_count	365	count
mean_component_length_px	87.6219	px
median_component_length_px	23	px
field_length_density_w_per_px	0.0360518	px^-1
field_density_without_flagged_per_px	0.0317114	px^-1
field_density_in_roi_per_px		px^-1
unattached_length_px	498	px
unattached_component_count	27	count
canalicular_length_density_attached_per_px	0.0314378	px^-1
field_length_density_with_unattached_per_px	0.0319273	px^-1
excluded_area_px2	0	px^2
analysed_field_area_px2	1.04858e+06	px^2
loop_count	130	count
loop_edge_share	0.2958	share
edge_count	1687	count

Canalicular width, the new measure

measure	value	units
width_mean_px	7.4669	px
width_median_px	8	px
width_p10_px	4	px
width_p90_px	10.198	px
width_pixels_used	30523	count

Canalicular width: 2 x the Euclidean distance transform of the canalicular mask at each skeleton pixel.
Measured only on skeleton pixels on the canalicular mask: the lacuna buffer and the
dilated vascular mask are excluded. The threads are about 2 to 6 px wide, so the width is close to the
pixel grid and quantised in steps of about 1 px, it moves with the threshold (a lower cut widens every
thread), and it has not been validated against any manual measurement.

Lacuna shape

lacuna_id	area_px2	major_axis_length_px	minor_axis_length_px	aspect_ratio	eccentricity	solidity	orientation_rad	centroid_row_px	centroid_col_px	on_border
1	954	58.9902	25.4792	2.3152	0.9019	0.6974	1.3025	49.6237	860.655	False
2	6236	133.392	66.6525	2.0013	0.8662	0.7565	0.7065	193.518	518.807	False
3	1740	108.483	22.6499	4.7895	0.978	0.7716	1.1221	244.43	785.516	False
4	4847	142.75	45.7551	3.1199	0.9472	0.8253	0.4966	381.669	319.923	False
5	2583	119.873	29.7486	4.0295	0.9687	0.7747	0.5429	394.401	41.6202	False
6	4698	104.479	62.6265	1.6683	0.8004	0.7983	0.887	422.75	620.834	False
7	2761	84.1795	44.2264	1.9034	0.8509	0.8577	1.3929	435.645	810.324	False
8	3441	99.3913	46.2527	2.1489	0.8851	0.8428	1.0959	709.919	822.946	False
9	2671	101.574	36.8708	2.7549	0.9318	0.8211	0.7949	796.647	571.938	False
10	594	35.3373	22.8099	1.5492	0.7638	0.8672	0.7727	907.293	697.131	False
11	740	61.1371	17.5677	3.4801	0.9578	0.8675	-1.5683	1016.29	147.323	True

Network per lacuna

lacuna_id	roots_count	ring_length_r30_px	ring_length_r60_px	owned_length_px	edge_count	mean_edge_length_px	ring_width_mean_r30_px
1	9	388	835	2379.41	96	24.7855	6.125
2	25	808	1745	5489.26	280	19.6045	7.2997
3	7	386	1023	2103.93	111	18.9543	7.2885
4	21	558	1326	3012.28	155	19.4341	7.3234
5	5	237	621	883.938	47	18.8072	5.4182
6	18	733	1545	3901.76	210	18.5798	7.2116

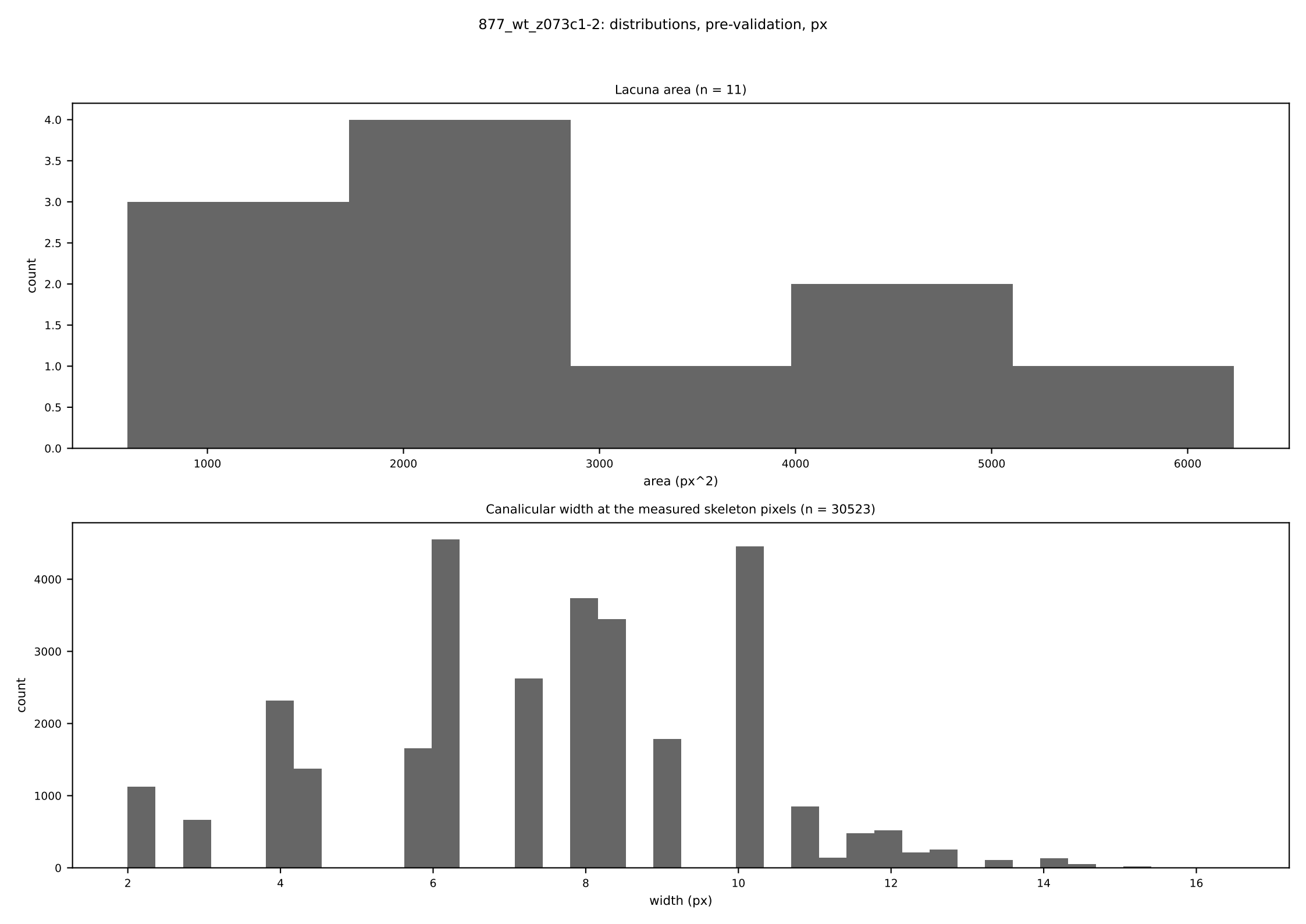
7	11	465	991	926.578	45	20.5906	6.5732
8	16	539	1091	1110.35	52	21.3528	5.9395
9	11	304	812	1385.25	70	19.7893	5.7254
10	0	121	334	235.48	10	23.548	7.125
11	0	0	0	0	0	0	

Normalised per lacuna

lacuna_id	perimeter_px	ring_area_r30_px2	ring_area_r60_px2	in_frame_fraction_r30	in_frame_fraction_r60	ring_density_r30	ring_density_r60	roots_per_100px_perimeter
1	218.244	7625	18201	1	0.8799	0.0508852	0.0458766	4.1238
2	510.192	15054	34735	1	1	0.0536734	0.0502375	4.9001
3	294.09	10047	25475	1	1	0.0384194	0.040157	2.3802
4	387.789	13352	32069	1	1	0.0417915	0.0413483	5.4153
5	312.576	9613	20973	0.8593	0.7542	0.0246541	0.0296095	1.5996
6	397.546	12204	28960	1	1	0.0600623	0.0533494	4.5278
7	261.764	9629	24171	1	1	0.0482916	0.0409995	4.2023
8	305.262	10522	26302	1	1	0.051226	0.0414797	5.2414
9	288.948	10418	25839	1	1	0.0291803	0.0314254	3.8069
10	124.024	5803	16722	1	1	0.0208513	0.0199737	0
11	159.918	3763	10281	0.5406	0.5269	0	0	0

Network v2 per lacuna

lacuna_id	ring_attached_length_r30_px	ring_attached_length_r60_px	ring_length_w_r30_px	ring_length_w_r60_px	sholl_crossings_r10	sholl_crossings_r20	sholl_crossings_r30
1	327	645	446.818	978.661	9	9	18
2	783	1683	911.654	2002.14	15	25	19
3	371	929	456.446	1188.67	9	11	17
4	485	1199	638.47	1541.63	12	15	22
5	157	330	269.179	708.169	7	5	14
6	677	1378	843.139	1774.78	18	22	26
7	464	924	540.63	1137.19	13	12	15
8	437	679	610.801	1242.76	9	14	16
9	255	508	345.262	920.465	11	7	8
10	64	91	131.498	373.877	5	4	5
11	0	0	0	0	0	0	0


877_wt_z073c1-2: distributions, pre-validation, px
Lacuna area (n = 11)
count
area (px^2)
Canalicular width at the measured skeleton pixels (n = 30523)
count
width (px)